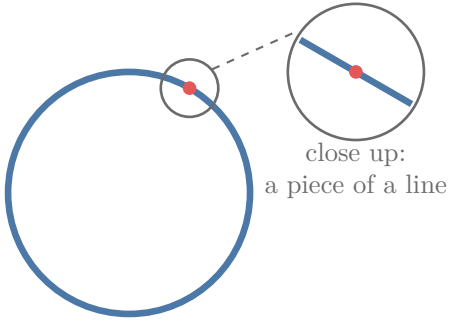
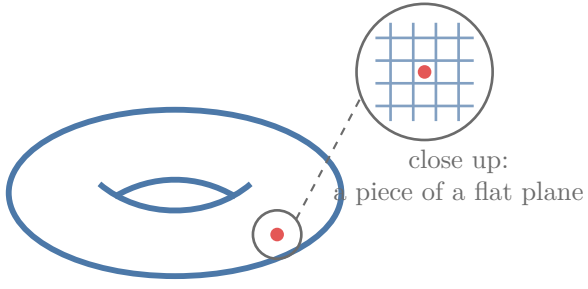
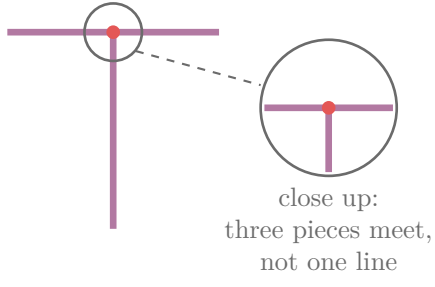
circle: a 1D manifold



torus: a 2D manifold



T-junction: not a manifold